VALUE runtime implementation overview

Executable market, storage, expansion, planning and fixed-zonal methods

Model clock	30-minute UTC periods; fixed 365-day model year; 17,520 periods
Core market	National bid at cost with current-period balancing
Annual chain	PSM, expansion headroom, owner investment, planning and state transition
Network option	Fixed-zonal transport and pay-as-bid redispatch
Document boundary	Edition 0.5; inputs, operating rules, annual accounts and network constraints

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Software methods reference, generated from the maintained source

VALUE 0.7.0-alpha.1 connects half-hourly electricity-system operation to annual investment, planning and asset retirement. The bilingual methodology, edition 0.5, gives the complete equations, numerical parameters, pseudocode and input tables. This overview identifies the runtime functions that perform those calculations.

1. Inputs and initial state

The model clock contains 17,520 UTC half-hours in a fixed 365-day year, with February represented by 28 days. Power uses MW, interval energy MWh and offers GBP/MWh. A complete operating year supplies the annual investment calculation.

native_initial_state reads fleet assets, REPD projects, technical costs, development timelines and regional success rates from the selected data pack. Assets carry capacity, technology, location, cost and economic life. Projects carry development status, capacity and commissioning year. Demand and import inputs are aligned to the model's UTC clock; the weather module converts wind and radiation to available generation.

The default configuration is value-corrected. value-bid-at-cost-psm uses the native-corrected-v1 rule set for national operation. The public2 data packs provide the operational inputs; the value-uk-research-suite-v1-public2 suite supplies the 23-zone network configuration.

2. Market operation and storage

National operation is bid at cost. Each half-hour schedules supply against the forecast and balances realised demand and availability. Offers are ordered by price rounded to £0.01/MWh, storage status and unrounded price. Accepted suppliers receive the stage's uniform marginal settlement price.

Thermal downward adjustments first use the output above the minimum stable level. A further shutdown segment compares avoided operating expenditure with restart expenditure over the estimated downtime. Recorded load shedding is valued at £17,000/MWh. The period balance also records additional physical shortfall; their sum enters annual unserved energy.

Battery.charge, Battery.discharge, Battery.buy_back and Battery.apply_self_discharge update stored energy, dated batches, charging and discharge losses, and self-discharge. Each asset has one net position per period and shares its rated power across clearing stages. The annual national adapter creates fresh storage objects at the beginning of each year.

The dynamic annual-average storage cost uses capital expenditure, fixed operating cost, economic life and the previous year's sold energy and dwell. DynamicAnnualStorageCost.prepare_year calculates annualised project cost and battery cycle depreciation. Default national bidding uses the battery cycle-depreciation component; pumped hydro and hydrogen have zero cycle-depreciation bids. The staged path evaluates bid_price_gbp_per_mwh at zero dwell.

3. Annual accounts, investment and planning

Annual accounts aggregate generation, storage sales, market revenue and physical costs. Gas and biomass investment profit deducts operating expenditure from income. Wind, solar and storage use gross revenue for the investment decision. Resource-cost accounting separately annualises capital expenditure with capital_recovery_factor.

vre-expansion-cap and value-storage-expansion-policy calculate technology expansion headroom from annual demand, generation profiles and storage use. The storage rule uses the surplus remaining after existing storage charging. Each battery type receives its own ceiling from this calculation.

agent-investment evaluates owner and technology-region groups using realised income, capital expenditure, preferred return and target payback. Its four decision tiers select investment, waiting or retirement. Development times and regional success rates come from the tables stored in the initial state.

The headroom calculation counts the operating fleet; pending projects retain their individual commissioning schedules.

The planning pipeline admits proposals as expected capacity or as seeded stochastic outcomes. planning.advance_year commissions due projects at the start of the year; planning.admit_projects admits newly proposed projects at year end. transition.apply advances asset age, applies retirement and constructs the next year's state. The previous year's storage-cost observations accompany this state.

4. Balancing and network constraints

value-staged-bid-at-cost-psm creates the national ahead schedule. value-copperplate-balancing balances a single node; value-zonal-redispatch-balancing applies fixed zonal network constraints and pay-as-bid redispatch. scenario_scaled_zonal_shares distributes the study demand across zones.

Zonal redispatch applies electricity balance, resource availability, storage bounds, directed corridor capacities and signed boundary constraints. The solver minimises four objectives in sequence: adjustment bid cost including shortage valuation; accepted upward and downward MWh plus shortage MWh; weighted physical throughput; and stable variable order. Each subsequent solve retains the preceding objective bounds within the calculated numerical tolerances.

The 23-zone suite assigns explicit locations to wind and solar. Other supplied resources use ENGLAND_FALLBACK, including the runtime nuclear station allocation and the country-level import assets. These assignments define the spatial input to the zonal calculation.

The DC network method calculates line power from voltage angles and branch reactance, with thermal capacity equal to the line rating times its circuit count. The AC calculation additionally solves voltage magnitudes, reactive-power balance and branch apparent-power limits. Chapter 7 specifies each formulation and its numerical stopping conditions.

5. Perfect foresight, hydrology and carbon

value-perfect-foresight-lp jointly optimises generation, imports, load shedding and storage over the declared time window. Storage inventory connects consecutive periods, and the terminal policy fixes or bounds its closing inventory. The independent functions run_of_river_dispatch and reservoir_dispatch calculate available run-of-river electricity and reservoir generation from inflows; reservoir dispatch also returns water storage, turbine release and spill.

build_operational_carbon_ledger multiplies generation and imported electricity by the selected emission factors. _asset_embodied_lines annualises construction emissions using power capacity or storage energy capacity and economic life. Total emissions combine these components; carbon intensity divides by served electricity. Chapter 9 gives the factors, units and storage-carbon inventory calculation.

6. Implementation locations

gridform_core/v2/orchestrator.py defines the annual sequence. canonical_psm_data.py constructs initial assets and projects; v2_module_definitions.py supplies the default annual investment and planning modules. cost_ledger.py and carbon_ledger.py aggregate economic and emissions accounts.

zonal_redispatch.py, zonal_contracts.py and zonal_solver_contract.py define network redispatch and numerical solution. Module declarations are in gridform_core/manifests. The bilingual chapters place the specific function names and input fields beside each calculation.